Table 5-6 Modeling Parameters and Acceptance Criteria for Nonlinear Procedures—Structural Steel Components (continued)

	Modeling Parameters			Acceptance Criteria				
	Plastic Rotation Angle, Radians		Residual Strength Ratio	Plastic Rotation Angle, Radians				
					Primary		Secondary	
Component/Action	**a**	**b**	**c**	**IO**	**LS**	**CP**	**LS**	**CP**
For $0.2 < P/P_{CL} < 0.50$								
a. $\frac{b_f}{2t_f} \le \frac{52}{\sqrt{F_{ye}}}$ and $\frac{h}{t_w} \le \frac{260}{\sqrt{F_{ye}}}$	—[3]	—[4]	0.2	$0.25\theta_y$	—[5]	—[3]	—[6]	—[4]
b. $\frac{b_f}{2t_f} \ge \frac{65}{\sqrt{F_{ye}}}$ or $\frac{h}{t_w} \ge \frac{400}{\sqrt{F_{ye}}}$	$1\theta_y$	$1.5\theta_y$	0.2	$0.25\theta_y$	$0.5\theta_y$	$0.8\theta_y$	$1.2\theta_y$	$1.2\theta_y$
c. Other	Linear interpolation between the values on lines a and b for both flange slenderness (first term) and web slenderness (second term) shall be performed, and the lowest resulting value shall be used							
Column Panel Zones	$12\theta_y$	$12\theta_y$	1.0	$1\theta_y$	$8\theta_y$	$11\theta_y$	$12\theta_y$	$12\theta_y$
Fully Restrained Moment Connections[13]								
WUF[12]	0.051-0.0013*d*	0.043-0.0006*d*	0.2	0.0128-0.0003*d*	0.0337-0.0009*d*	0.0284-0.0004*d*	0.0323-0.0005*d*	0.043-0.0006*d*
Bottom haunch in WUF with slab	0.026	0.036	0.2	0.0065	0.0172	0.0238	0.0270	0.036
Bottom haunch in WUF without slab	0.018	0.023	0.2	0.0045	0.0119	0.0152	0.0180	0.023
Welded cover plate in WUF[12]	0.056-0.0011*d*	0.056-0.0011*d*	0.2	0.0140-0.0003*d*	0.0319-0.0006*d*	0.0426-0.0008*d*	0.0420-0.0008*d*	0.056-0.0011*d*
Improved WUF-bolted web[12]	0.021-0.0003*d*	0.050-0.0006*d*	0.2	0.0053-0.0001*d*	0.0139-0.0002*d*	0.0210-0.0003*d*	0.0375-0.0005*d*	0.050-0.0006*d*
Improved WUF-welded web	0.041	0.054	0.2	0.0103	0.0312	0.0410	0.0410	0.054
Free flange[12]	0.067-0.0012*d*	0.094-0.0016*d*	0.2	0.0168-0.0003*d*	0.0509-0.0009*d*	0.0670-0.0012*d*	0.0705-0.0012*d*	0.094-0.0016*d*
Reduced beam section[12]	0.050-0.0003*d*	0.070-0.0003*d*	0.2	0.0125-0.0001*d*	0.0380-0.0002*d*	0.0500-0.0003*d*	0.0525-0.0002*d*	0.07-0.0003*d*
Welded flange plates								
a. Flange plate net section	0.03	0.06	0.2	0.0075	0.0228	0.0300	0.0450	0.06
b. Other limit states	force-controlled							
Welded bottom haunch	0.027	0.047	0.2	0.0068	0.0205	0.0270	0.0353	0.047
Welded top and bottom haunches	0.028	0.048	0.2	0.0070	0.0213	0.0280	0.0360	0.048
Welded cover-plated flanges	0.031	0.031	0.2	0.0078	0.0177	0.0236	0.0233	0.031